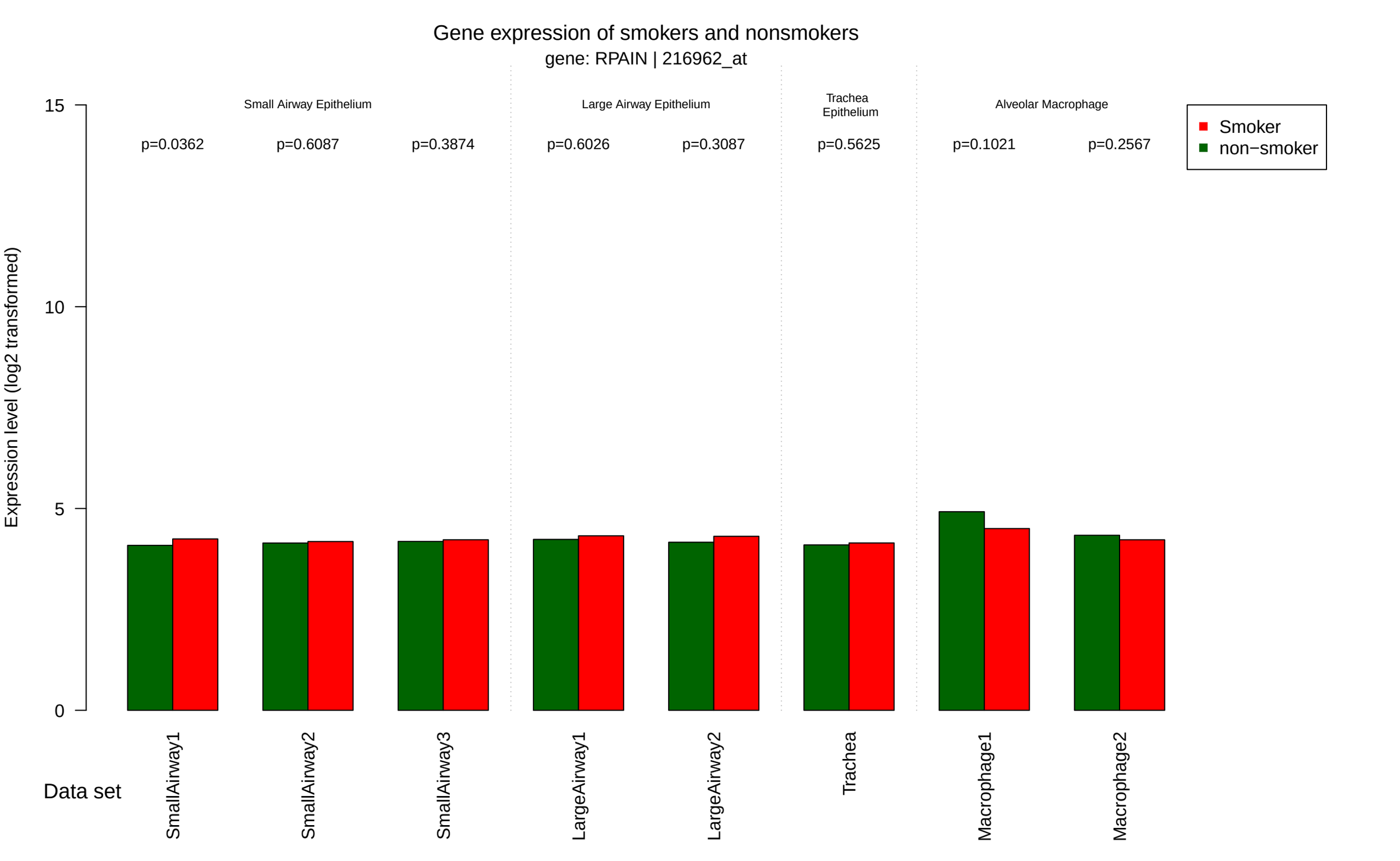
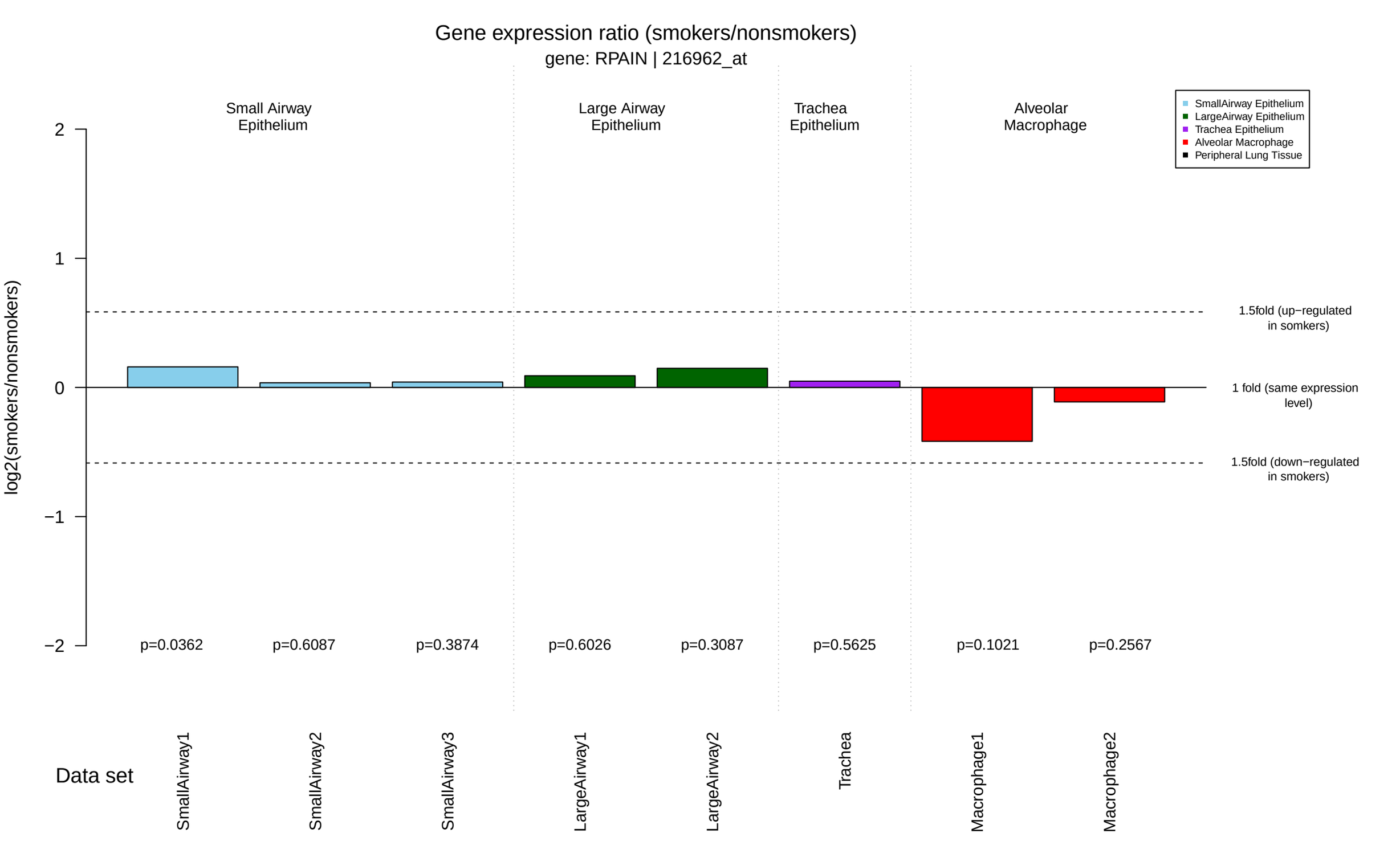
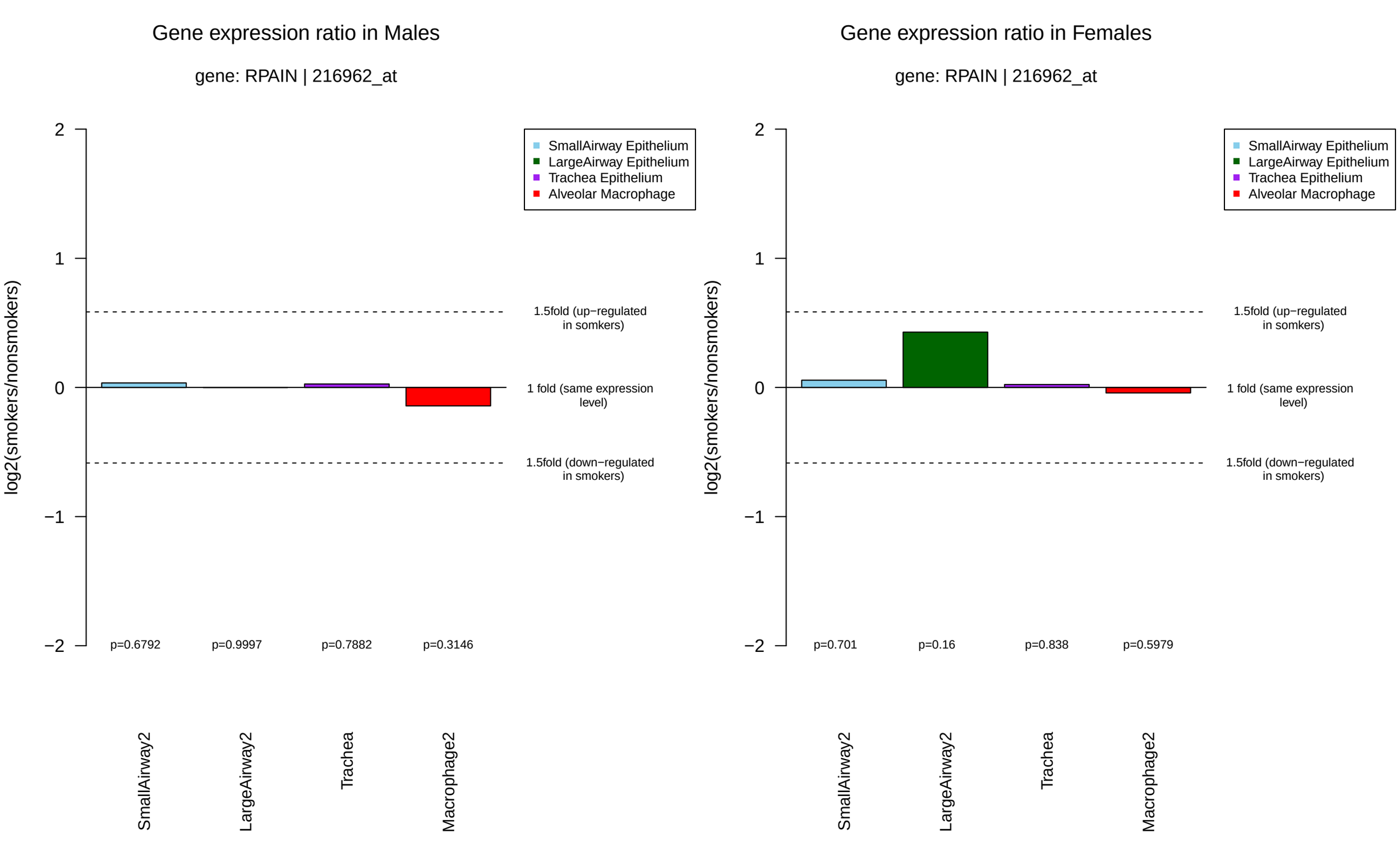
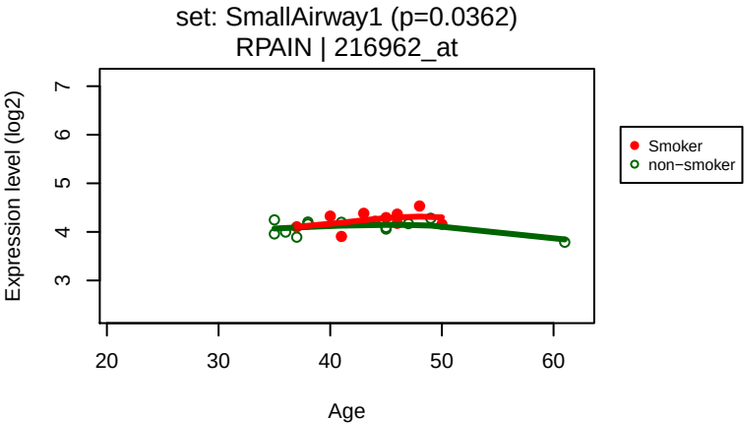
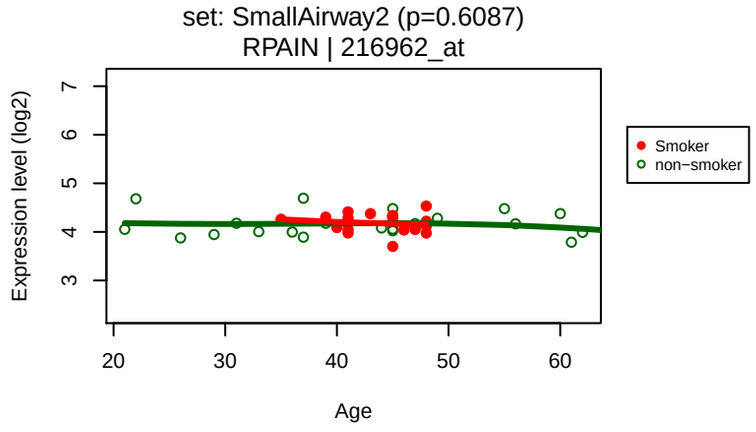
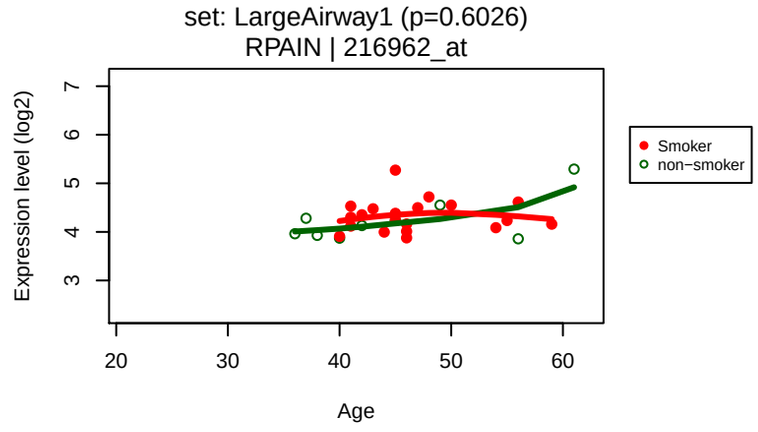

Gene expression of smokers and nonsmokers
gene: RPAIN | 216962_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0362
p=0.6087
p=0.3874
p=0.6026
p=0.3087
p=0.5625
p=0.1021
p=0.2567
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

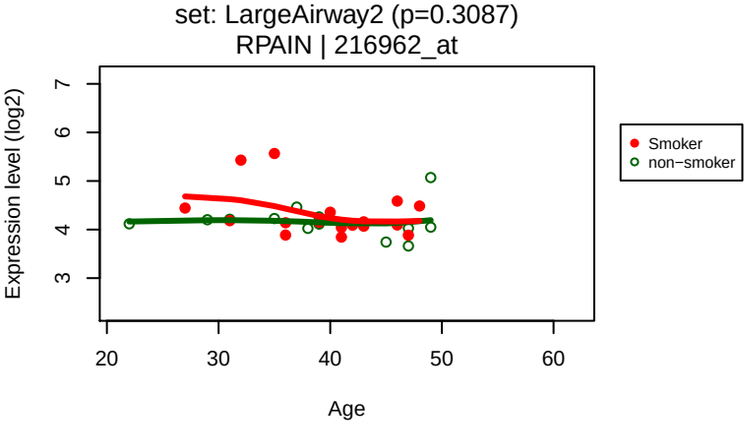

Gene expression ratio (smokers/nonsmokers)
gene: RPAIN | 216962_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0362
p=0.6087
p=0.3874
p=0.6026
p=0.3087
p=0.5625
p=0.1021
p=0.2567
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

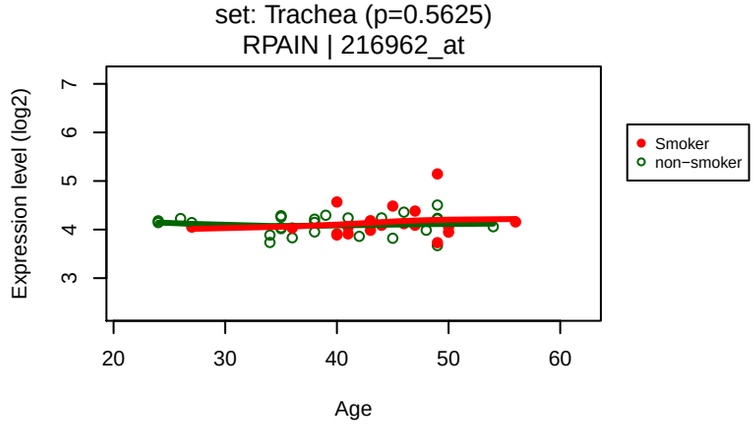

Gene expression ratio in Males
gene: RPAIN | 216962_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6792
p=0.9997
p=0.7882
p=0.3146
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPAIN | 216962_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.701
p=0.16
p=0.838
p=0.5979
SmallAirway2
LargeAirway2
Trachea
Macrophage2

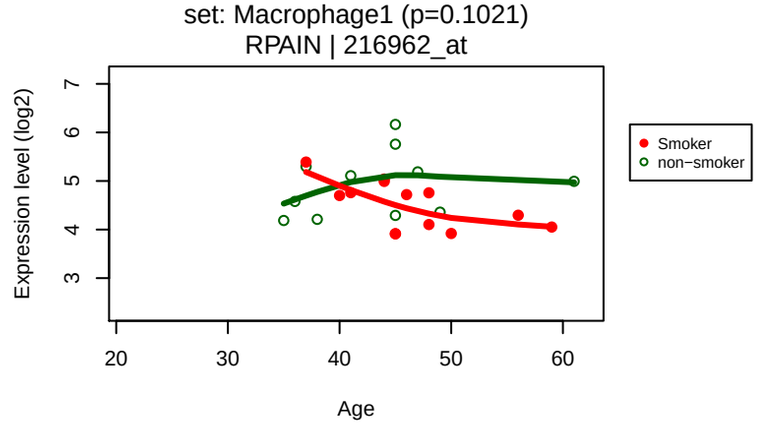

set: SmallAirway1 (p=0.0362)
RPAIN | 216962_at
Expression level (log2)
Age
Smoker
non−smoker

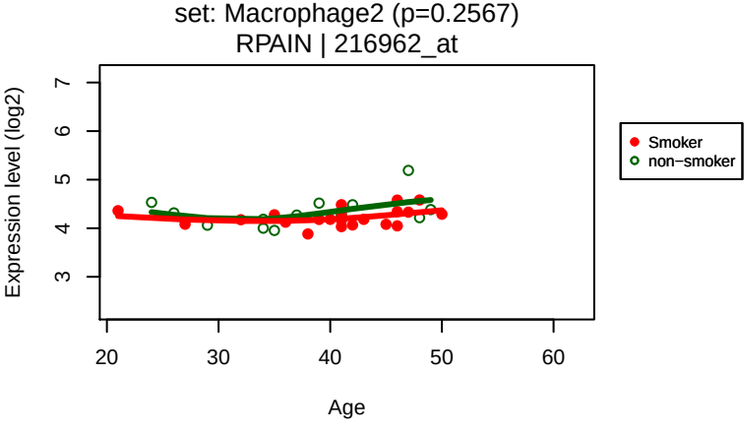

set: SmallAirway2 (p=0.6087)
RPAIN | 216962_at
Expression level (log2)
Age
Smoker
non−smoker

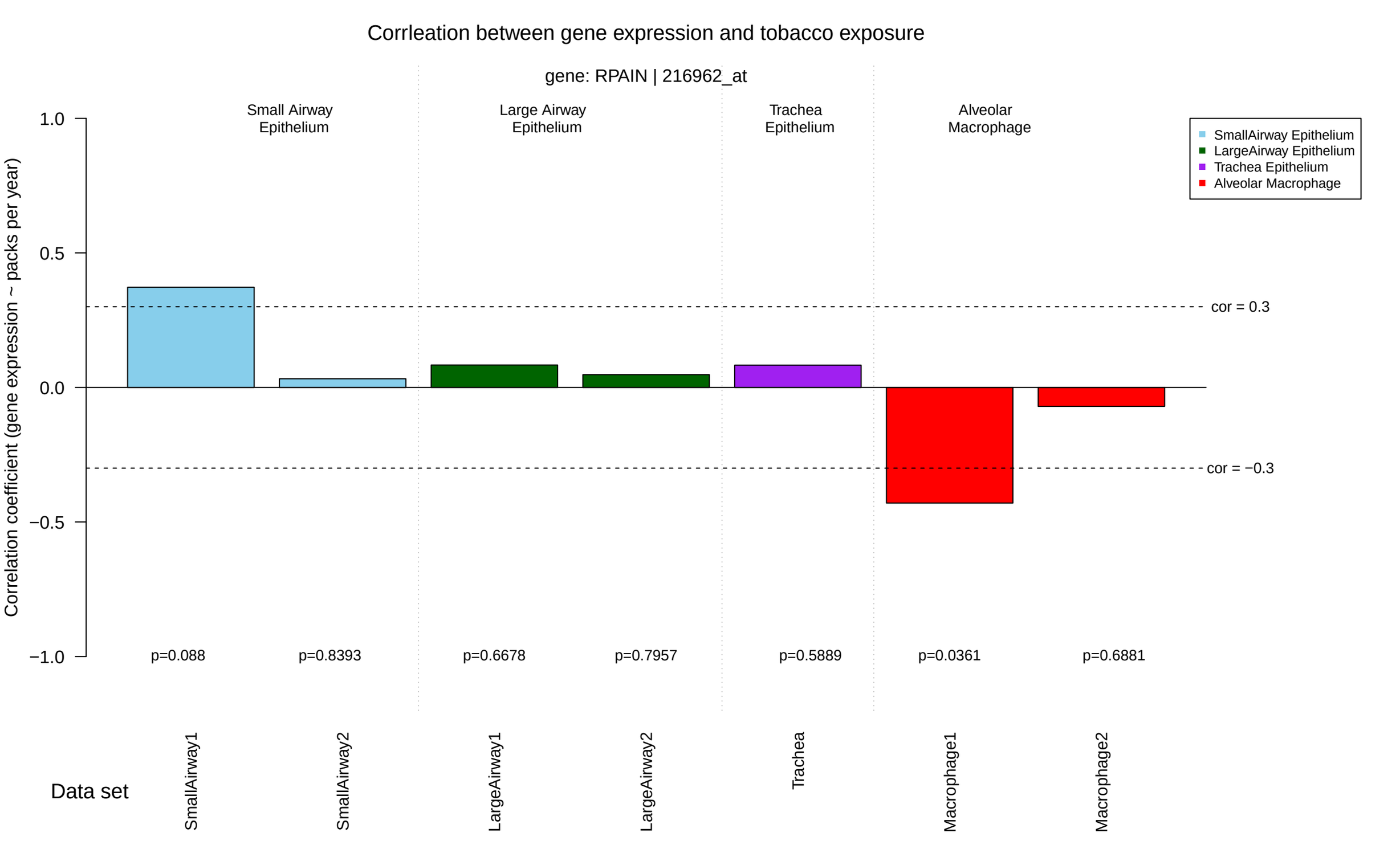

set: LargeAirway1 (p=0.6026)
RPAIN | 216962_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3087)
RPAIN | 216962_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5625)
RPAIN | 216962_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1021)
RPAIN | 216962_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2567)
RPAIN | 216962_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPAIN | 216962_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.088
p=0.8393
p=0.6678
p=0.7957
p=0.5889
p=0.0361
p=0.6881
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2